

Ryan M. C. Law

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Education

<i>PhD in Medical Science</i> MRC Cognition and Brain Sciences Unit, University of Cambridge, UK Supervisors: Dr Olaf Hauk & Prof Matthew A. Lambon-Ralph	2022 – 2026
<i>MSc in Cognitive Neuroscience (Research)</i> Donders Institute for Brain, Cognition and Behaviour, Radboud University, NL Supervisors: Dr Hugo Weissbart & Dr Andrea E. Martin	2021 – 2022
<i>BA in Linguistics</i> University College London, UK Supervisor: Dr Wing Yee Chow	2015 – 2018

Awards and Scholarships

<i>Travel Grant</i> , 20th Alpine Brain Imaging Meeting (CHF1,000)	2026
<i>Travel Grant</i> (shortlisted), Society for the Neurobiology of Language	2025
<i>Diverse Intelligences Summer Institute Project Accelerator Grant</i> (\$3,000)	2024
<i>Student Award</i> , 23rd International Conference on Biomagnetism (AUD500)	2024
<i>Diverse Intelligences Summer Institute Fellow</i> , St Andrews, UK	2024
<i>Gates Cambridge Scholarship</i> (Full PhD funding at Overseas rate, ~£250k), accepted	2022
<i>Leverhulme Doctoral Training Programme for the Ecological Study of the Brain</i> , declined	2022
<i>UCL Overseas Research Scholarship</i> (PhD tuition at Overseas rate), declined	2022
<i>UCL Graduate Research Scholarship</i> (PhD tuition at UK Home rate and maintenance), declined	2022
<i>UCL-Birkbeck Medical Research Council Doctoral Training Programme Funding</i> (PhD tuition at UK Home rate and maintenance), declined	2022
<i>UCL International Scholar Award for Doctoral Training</i> (PhD tuition at Overseas rate), declined	2022
<i>Featured Scientist</i> , 100 Scientists of Malaysia, Malaysia	2021
<i>Best Presentation Prize</i> , UCL Faculty of Brain Sciences Undergraduate Conference	2018
<i>Sultan Ibrahim Johor Bursary</i> , State of Johor, Malaysia. (Awarded to promising Malaysian students studying abroad from the Sultanate of Johor in Malaysia).	2018
<i>Summer Undergraduate Research Internship Bursary</i> , Harvard University, USA.	2017
<i>UCL International Short Mobility Bursary</i>	2017
<i>Award for Academic Excellence</i> , UCL Department of Linguistics, UK.	2015

Publications

Peer-reviewed articles

- [1] **Law, R.M.C.**, Lambon Ralph, M.A., Hauk, O. (2026). A common framework for semantic memory and semantic composition. *Imaging Neuroscience*.
- [2] Rane, S. *, Kirkman, C.F. *, Todd, G. *, Royka, A. *, **Law, R.M.C.** *, Cartmill, E., Foster, J.G. (*accepted*). Position: Principles of Animal Cognition to Improve LLM Evaluations. *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. (*equal contribution.)
- [3] Hauk, O., Magnabosco F., **Law, R.M.C.** (2023). Can we separate semantic representations from computations? A commentary on Calzavarini (2023). *Language, Cognition and Neuroscience*.
- [4] **Law, R.**, Pykkänen, L. (2021). Lists with and without syntax: A new approach to measuring the neural processing of syntax. *Journal of Neuroscience*.
- [5] Tulling, M., **Law, R.**, Cournane, A., Pykkänen, L. (2020). Neural correlates of modal displacement and discourse-updating under (un)certainly. *eNeuro*.

Manuscripts in preparation

- [1] **Law, R.M.C.**, Rogers, T.T., Lambon Ralph, M.A. (*in prep*). A dual-stream neural network model of semantic cognition.
- [2] **Law, R.M.C.**, Hauk, O., Lambon Ralph, M.A., (*in prep*). From *Bird* to *Nocturnal Bird* to *Owl*: MEG-EEG insights into anterior temporal lobe function in semantic processing.
- [3] **Law, R.**, Kaufeld, G., Bosker, H.R., Martin, A.E., (*in prep*). Cortical tracking of linguistic units at different speech rates.
- [4] **Law, R.**, Weissbart, H., Martin, A.E., (*in prep*). Representational dynamics of meaning composition during natural listening.

Presentations

Invited talks

- [1] *University of Geneva, NeuroLingo Lab*. Geneva, Switzerland. (2026, January)
- [2] *University of Oxford, Psycholinguistics Seminar*. Oxford, UK. (2025, May)
- [3] *New York University, Neuroscience of Language Lab*, USA. (2025, May)
- [4] *University of Cambridge, Cambridge Neuroscience*, Cambridge, UK. (2025, March).
- [5] *Norwich Science Festival*, Norwich, UK. (2025, February).
- [6] *UCL, Linguistics Departmental Seminar Series*. London, UK. (2024, May).
- [7] *Bristol University, Department of Computer Science*. Bristol, UK. (2023, December).
- [8] *Dartmouth College, Department of Linguistics*. Hanover, New Hampshire, USA. (2020, October).

Slide presentations

- [1] **Law, R.M.C.**, Hauk, O., Lambon Ralph, M.A. (2026, January). Single word conceptual specificity, not phrasal composition, drives anterior temporal lobe during semantic processing: evidence from combined EEG/MEG. Blitz talk presentation at the 20th Alpine Brain Imaging Meeting 2026. Champéry, Switzerland.

- [2] **Law, R.M.C.**, Lambon Ralph, M.A., Hauk, O. (2025, September). A common framework for semantic memory and semantic composition. Slide presentation at the Society for the Neurobiology of Language 17th Annual Meeting. Washington DC, USA.
- [3] **Law, R.M.C.**, Hauk, O., Lambon Ralph, M.A. (2025, July). Meaning construction in the anterior temporal lobe: an MEG/EEG study. Blitz talk presentation at the MEG-UKI Conference 2025. London, UK.
- [4] Rane, S.*, Kirkman, C.F.*, Todd, G.*, Royka, A.*, **Law, R.M.C.***, Cartmill, E., Foster, J.G. (2024, December). Principles of Animal Cognition for LLM Evaluations: A Case Study on Transitive Inference. *NeurIPS 2024 Workshop on Behavioral Machine Learning*. Vancouver, Canada. (*equal contribution.)
- [5] **Law, R.M.C.**, Lambon Ralph, M.A., Hauk, O. (2024, August). Tracking compositional semantic processing using MEG and EEG. Slide presentation at the 23rd International Conference on Biomagnetism. Sydney, Australia.
- [6] **Law, R.**, Weissbart, H., Martin, A.E. (2022, May). Decoding composition during natural listening. Slide presentation at the Radboud University Cognitive Neuroscience Conference Synapsium. Nijmegen, the Netherlands.
- [7] **Law, R.**, Kaufeld, G., Bosker, H.R., Martin, A.E. (2021, September). Cortical tracking of linguistic units at different rates. Slide slam at 13th Annual Meeting of the Society for the Neurobiology of Language Virtual Edition.
- [8] **Law, R.**, Pylkkänen, L. (2020, May). Lists with and without syntax: Neural correlates of syntactic structure. Data blitz presentation at the CNS 2020 Virtual Meeting.
- [9] **Law, R.**, Pylkkänen, L. (2019, April). Neural correlates of syntactic structure: MEG study using lists. Slide presentation at the Neuroscience of Language Conference 2019. Abu Dhabi, UAE.
- [10] **Law, R.**, Arkhipova, Y., Chow, W.Y. (2018, March). Tracking expectation reversal during discourse comprehension using EEG. Slide presentation at the Faculty of Brain Sciences Fifth Annual Undergraduate Conference. London, UK.

Poster presentations

- [1] **Law, R.M.C.**, Hauk, O., Lambon Ralph, M.A. (2026, January). Single word conceptual specificity, not phrasal composition, drives anterior temporal lobe during semantic processing: evidence from combined EEG/MEG. Poster presentation at the 20th Alpine Brain Imaging Meeting 2026. Champéry, Switzerland.
- [2] **Law, R.M.C.**, Hauk, O., Lambon Ralph, M.A. (2025, July). Meaning construction in the anterior temporal lobe: an MEG/EEG study. Poster presentation at the MEG-UKI Conference 2025. London, UK.
- [3] **Law, R.M.C.**, Lambon Ralph, M.A., Hauk, O., (2024, October). Tracking the neural dynamics of compositional semantics with MEG and EEG. Poster presentation at the MEG-UKI Conference 2024. Birmingham, UK.
- [4] **Law, R.M.C.**, Lambon Ralph, M.A., Hauk, O., (2023, October). An MEEG study bridging semantic composition and semantic cognition. Poster presentation at the 15th Annual Meeting of the Society of the Neurobiology of Language. Marseille, France.
- [5] **Law, R.**, Weissbart, H., Martin, A.E., (2022, June). Decoding composition during natural listening. Poster presentation at the International Max Planck Research School Conference: Interdisciplinary Approaches to the Language Sciences. Nijmegen, the Netherlands.
- [6] **Law, R.**, Kaufeld, G., Bosker, H.R., Martin, A.E., (2022, April). Cortical tracking of linguistic units at different rates. 18th NVP Winter Conference on Brain and Cognition. Egmond aan Zee, the Netherlands.

- [7] **Law, R.**, Pykkänen, L. (2020, June). Lists with and without syntax: Neural correlates of syntactic structure. Virtual poster presentation at the International Max Planck Research School Conference: Interdisciplinary Approaches to the Language Sciences.
- [8] **Law, R.**, Pykkänen, L. (2020, May). Lists with and without syntax: Neural correlates of syntactic structure. Poster presentation at the CNS 2020 Virtual Meeting.
- [9] **Law, R.**, Pykkänen, L. (2019, October). Lists with and without syntax: MEG effects of syntactic composition when local conceptual combination is controlled. Poster presentation at the International Brain and Syntax Think Tank. Evanston, USA.
- [10] Arkhipova, Y., Sostarics, T., **Law, R.**, Xiang, M., Chow, W.Y. (2019, September). ‘Nonetheless’ can reverse predictions immediately: evidence from ERPs. Poster presentation at the 25th Architectures and Mechanisms of Language Processing Conference. Moscow, Russia.
- [11] **Law, R.**, Pykkänen, L. (2019, August). Lists with and without syntax: MEG effects of syntactic composition when local conceptual combination is controlled. Poster presentation at the 11th Annual Meeting of the Society for the Neurobiology of Language. Helsinki, Finland.
- [12] Tulling, M., **Law, R.**, Cournane, A., Pykkänen, L. (2019, August). Discourse belief updating in the right hemisphere. Poster presentation at the 11th Annual Meeting of the Society for the Neurobiology of Language. Helsinki, Finland.
- [13] Arkhipova, Y., **Law, R.**, Xiang, M., Chow, W.Y. (2019, March). ‘Nonetheless’ can reverse predictions immediately: evidence from ERPs. Poster presentation at the 32nd Annual CUNY Human Sentence Processing Conference. Boulder, USA.

Teaching, supervision & mentorship

<i>Supervisor</i> , Psychology of Language Processing & Learning, University of Cambridge	2024-2025
<i>Mentor</i> , Cambridge Access to Mental Health & Psychology, Department of Psychiatry, University of Cambridge (aimed at graduate school applicants from underrepresented backgrounds in Psychology and Psychiatry)	2022
<i>Guest lecturer</i> , Psycholinguistics, Dartmouth College, USA	2020
<i>Instructor</i> , MEG preprocessing and analysis workshop, New York University Abu Dhabi, UAE	2020

Service

<i>Ad-hoc reviewer</i> for Language, Cognition, and Neuroscience, Topics in Cognitive Science, Journal of Psycholinguistic Research, Cognitive Science Conference	
<i>Program Committee member</i> , 18 th Annual Meeting of the Society for the Neurobiology of Language	2026
<i>Elected student and post-doc representative to the Board</i> , Society for the Neurobiology of Language	2025-present
<i>Equity, Diversity, and Inclusion Committee member</i> , MRC Cognition and Brain	2023–present

Science Unit, University of Cambridge

Diversity, Equity, and Inclusion Committee member, Max Planck Institute for Psycholinguistics 2020–2022

Conference organising committee, Neuroscience of Language Conference Abu Dhabi 2019

Skills

Programming languages: Python (7 years), R, MATLAB, JavaScript

Natural languages: English • Mandarin • Malay • Cantonese • Teochew • Hokkien • Dutch (B1) • Arabic (A2)

Research positions

Research project coordinator, Max Planck Institute for Psycholinguistics 2020 – 2022

Lab rotation student, multiple labs at Radboud University 2021 – 2022
Exploring transcranial ultrasound stimulation, neurobiology of language

Predoctoral researcher, New York University 2018 – 2020

Research assistant/intern, multiple labs & institutions (Harvard, UCL) 2016-2018
Exploring the psychology, philosophy, neuroscience, and development of language